Returns to Scale

LLE Mathematics and Statistics

For each of the following production functions, identify the returns to scale.

1. $Q(K, L) = 3K + 2L$
2. $Q(K, L) = 0.2KL$
3. $Q(K, L) = \frac{KL}{2}$
4. $Q(K, L) = K^2L^2$
5. $Q(K, L) = K^{0.5}L^{0.5}$
6. $Q(K, L) = K^{0.9}L^{0.1}$
7. $Q(K, L) = K^{0.2}L^{0.6}$
8. $Q(K, L) = K^{0.8}L^{0.3}$
9. $Q(K, L) = K^{\alpha}L^{\beta}$ where
   (a) $\alpha + \beta = 1$
   (b) $\alpha + \beta < 1$
   (c) $\alpha + \beta > 1$